MILITARY SPECIFICATION

TEST SET, RADIO TS-1588( )/AIC

1. SCOPE

1.1 This specification covers one type of test set which provides facilities for checking the serviceability of Control, Intercommunication Set C-1611( )/AIC and is designated Test Set, Radio TS-1588( )/AIC (See 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid, or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS:

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, Double Faced, Stringer Construction
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Non Metallic (and Connectors)

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PPP-T-76	Tape, Pressure-Sensitive, ADHESIVE PAPER (For Carton Sealing)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced
QQ-S-781	Strapping, Steel, Flat and Seals

MILITARY

MIL-F-14072	Finish for Ground Signal Equipment
MIL-M-13231	Marking of Electronic Items
MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

USAECOM

DL SM-B-317130 - Test Set, Radio TS-1588( )/AIC Assembly

(Copies of specifications, standards, and drawings required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Description. - Test Set, Radio TS-1588( )/AIC contains switches, cables, and connectors to facilitate the interconnection and control of test equipment for bench testing the circuitry of an airborne interphone control unit (Control, Intercommunication Set C-1611( )/AIC.)

3.2 Construction. - The Test Set shall be constructed in accordance with Data List No. DL SM-B-317130 - Test Set, Radio TS-1588( )/AIC Assembly.

3.3 Preproduction units. - The contractor shall furnish the number of preproduction samples as specified by the invitation for bid and contract (See 4.3 and 6.2).

3.4 Cleaning. -

3.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish, protective. - The equipment shall be given protective finish in accordance with MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touchup after mounting. The final paint film on Type 1 surfaces shall be Film E in accordance with MIL-F-14072 (See 4.4).

3.7 Securing of parts. -

3.7.1 General. - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7.2 Securing of parts by threaded fasteners. - Nylok or other screws with plastic devices and Loctite or similar sealants shall not be subjected to temperature in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

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3.8 Marking. -

3.8.1 General. - Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification. (See 4.4)

3.8.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.8.3 Serial numbers. - The following items shall be serial numbered:

Test Set, Radio TS-1588( )/AIC
Panel Assembly, Test Set SB-( )
Cover, Test Set CW-( )

3.9 Soldering. -

3.9.1 Solder. - Solder shall conform to the equipment drawings.

3.9.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area, or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and Cabling. - Wiring and cabling shall be neat and sturdy.

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3.10.1 Cabling. - Insulated wires shall be formed into cables, except where operation of the equipment would be adversely affected thereby, or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance.

3.10.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the parts to which the wire or cable is connected, except where un-insulated wires are used.

3.10.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.10.4 Support. - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable during and after subjection of the equipment to specified service conditions or after service or repair of the equipment in a normal manner.

3.10.5 Clearance. - Clearance between solder connections or bare connectors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.10.7 Connections. -

3.10.7.1 General. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

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3.11 Tropicalization of materials. - Materials shall be treated as follows:

3.11.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treated material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated materials will be held to the practical minimum.

3.11.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.11.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11.4 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.

3.12 Service conditions. (See 4.8) - TS-1588( )/AIC shall meet the following service conditions. Where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.12.1 Temperature (See 4.8.1). -

3.12.1.1 Operating. - The TS-1588( )/AIC shall operate satisfactorily within a temperature range of 0° to +55°C.

3.12.1.2 Non-operating. - The TS-1588( )/AIC shall operate satisfactorily following storage temperature of -62°C to +71°C.

3.12.2 Humidity (See 4.8.2). - The TS-1588( )/AIC shall operate satisfactorily in humidity of 94 ± 4 percent (94 ± 4%).

3.12.3 Altitude (See 4.8.3). -

3.12.3.1 Operating. - The TS-1588( )/AIC shall operate satisfactorily in altitude conditions up to 10,000 feet above sea level.

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3.12.3.2 Non-operating. - The TS-1588( )/AIC shall operate satisfactorily after being subjected to altitude conditions up to 50,000 feet above sea level.

3.12.4 Spray-tight protection (See 4.8.4). - The equipment shall operate satisfactorily after being subjected to rain, experienced during normal service conditions.

3.12.5 Fungus (See 4.8.5). - The unit(s) shall show no evidence of viable fungus after being subjected to 28 days of fungus test.

3.12.6 Vibration (See 4.8.6). The TS-1588( )/AIC shall operate satisfactorily after being subjected to vibration conditions experienced during normal handling and transportation.

3.12.7 Shock (See 4.8.7). - The TS-1588( )/AIC shall operate satisfactorily after being subjected to shock conditions experienced during normal handling and transportation.

TABLE I

Paragraph Heading	Paragraph Number	Performance After Test
Shock bench handling	4.8.7.1	(Note A) No physical damage, operable.
Shock drop	4.8.7.2	(Note A) Any physical damage shall be minor only.

NOTE A: The equipment shall meet the requirements listed in 3.13.

3.13 Electrical (See 4.9). -

3.13.1 Operational self check (See 4.9.1). - With 27.5 ± 0.5 volt dc applied and Power-Off switch S7 in power position, the test set shall meet the requirements as indicated in Table II.

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TABLE II

SWITCH FUNCTION	SWITCH POSITIONS									LIGHT INDICATIONS		
	S1	S2	S3	S4	S5	S6	I1	I2	I3	I1	I2	I3
1	Mid	R	R	Off	Off	SM	P	-	P	(X)	X	(X)
2	Radio	R	R	Off	1	SM	-	-	-	0	X	X
3	Mid	R	R	Off	1	SM	-	-	-	0	X	0
4	Inter	R	R	Off	1	SM	-	-	-	0	X	X
5	Radio	R	R	Off	2	SM	-	-	-	0	X	X
6	Mid	R	R	Off	2	SM	-	-	-	0	X	0
7	Inter	R	R	Off	2	SM	-	-	-	0	X	X
8	Mid	R	R	Off	2	SM	P	-	P	(X)	X	(X)
9	Mid	R	R	Off	3	SM	P	-	P	(X)	X	(X)
10	Radio	R	R	Off	3	SM	-	-	-	0	X	X
11	Mid	R	R	Off	3	SM	-	-	-	0	X	0
12	Inter	R	R	Off	3	SM	-	-	-	0	X	X
13	Mid	R	R	Off	4	SM	P	-	P	(X)	X	(X)
14	Radio	R	R	Off	4	SM	-	-	-	0	X	X
15	Mid	R	R	Off	4	SM	-	-	-	0	X	0
16	Inter	R	R	Off	4	SM	-	-	-	0	X	X
17	Mid	R	R	Off	ALTN	SM	P	-	P	(X)	X	(X)
18	Radio	R	R	Off	ALTN	SM	-	-	-	0	X	X
19	Mid	R	R	Off	ALTN	SM	-	-	-	0	X	0
20	Inter	R	R	Off	ALTN	SM	-	-	-	0	X	X
21	Mid	R	R	Off	ALTN	Hd Mic & Release	-	-	-	0	X	(X)
22	Mid	T	R	Off	4	"	-	-	-	0	X	(X)
23	Mid	T	R	Off	3	"	-	-	-	0	X	(X)
24	Mid	T	R	Off	2	"	-	-	-	0	X	(X)
25	Mid	T	R	Off	1	"	-	-	-	0	X	(X)
26	Mid	T	R	1	1	SM	-	-	-	0	X	X
27	Mid	T	R	2	1	SM	-	-	-	0	X	X
28	Mid	T	R	3	1	SM	-	-	-	0	X	X
29	Mid	T	R	4	1	SM	-	-	-	0	X	X
30	Mid	T	R	5	1	SM	-	-	-	0	X	X
31	Mid	T	R	6	1	SM	-	-	-	0	X	X
32	Mid	T	R	7	1	SM	-	-	-	0	X	X
33	Mid	T	R	INTER	1	SM	-	-	-	0	X	X
34	Mid	T	R	EMER	1	SM	-	-	-	0	X	X
35	Mid	T	R	Off	1	SM	-	-	-	0	X	0
36	Mid	R	T	Off	1	SM	-	-	-	0	X	0
37	Mid	I	I	Off	1	SM	-	-	-	0	X	X
38	Mid	I	I	Off	Off	SM	-	-	-	0	X	0

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TABLE II (Cont'd) -

Switch Function 39: Repeat function 3 thru 21, except that S2 shall be in position I and then position C.

NOTES:

X	-	Light On
(X)	-	Light On and Extinguish
O	-	Light Does Not Illuminate
P	-	Press & Release
Mid	-	Mid Position
SM	-	Signal Meter
-	-	Do Not Press

3.13.2 Posts, binding (J5) and (J6), (See 4.9.2). - With multimeter connected to +28V (J5) and GND (J6) binding posts, and POWER OFF switch (S7) set to POWER, the multimeter shall read 27.5 ± 0.5 volts dc.

3.13.3 Posts, binding (J2) and (J3), (See 4.9.3). - With no power applied to test set, and multimeter connected to AUDIO OSC-GND (J2) and OUTPUT MTR-GND (J3) binding posts, the resistance shall not exceed 0.5 ohms.

3.13.4 Post, binding (J1), (See 4.9.4). - With no power applied to the test set, S4 in 1 position, S3 in R position, and multimeter connected to pin 30 of connector (P1) and AUDIO OSC binding post (J1), the resistance shall not exceed 0.5 ohms.

3.13.5 Post, binding (J4), (See 4.9.5). - With no power applied to the test set, S2 in R position, and multimeter connected to pin 5 of connector (P1) and OUTPUT MTR binding post (J4), the resistance shall not exceed 0.5 ohms.

3.13.6 Jack (J7), (See 4.9.6). - With no power applied to the test set and plug (U-174/U) inserted in HDST MIC Jack (J7), and one lead of multimeter connected to pins 3, 5, 22 and 23 (S6 in HD Mic position) of connector (P1), and other to pins 3, 2, 1 and 4 respectively of jack, the resistance shall not exceed 0.5 ohms.

3.14 Bounce preconditioning (See 4.6 and 4.5). - The equipment shall meet the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning.

3.15 Operational inspection (See 4.12). - The test sets shall meet the specified test, after applicable processes, inspection and tests are completed, to determine that each test set on order is operable prior to packaging.

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3.16 Interchangeability (See 4.11). - Like units, assemblies, sub-assemblies, modules and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance; however, matched pairs of sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.17 Technical literature and spare parts. - Technical literature and running spares and spare parts shall be furnished if specified in the contract. Spare parts shall be identical to corresponding parts in the equipment furnished on the order (See 6.2).

3.18 Workmanship (See 4.10). - The equipment shall be manufactured and assembled in accordance with the equipment drawings and requirement nine (9) of MIL-STD-454 and the following paragraphs herein:

Cleaning	3.4
Controls	3.5
Securing of parts	3.7
Soldering	3.9
Wiring and Cabling	3.10
Tropicalization of materials	3.11

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that suppliers and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery) (See 4.3).

b. Inspection covered by subsidiary documents (See 4.4).

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c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.13 and 4.14).

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A, Group B and Group C, (See Tables III, IV and V respectively). The preproduction inspection will normally be performed in this order: (1) Altitude, (2) temperature, (3) vibration, (4) humidity, (5) shock, (6) shock, bench handling and (7) shock, drop. Other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of the units before preparation for delivery:

ITEM	WHERE REQUIRED
Finish	3.6
Marking	3.8

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.6. This does not relieve the contractor of his responsibility for performance of any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (See 6.7). Test equipment for Government verification inspection shall be made available by the contractor. Each unit shall be bounce preconditioned after final assembly and this preconditioning shall not be considered as a portion of acceptance inspection (See 3.14).

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4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for compliance to all the inspection and test requirements of Table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105, using general inspection level I and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection 4.12 shall be last (See 6.6).

TABLE III - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.18	4.10	0.65%	2.5%
Cable Assemblies (Assembled)			1.0%	4.0%
Cover CW-( )			1.0%	4.0%
Panel Assembly, SB-( )			1.0%	4.0%
Test Set, TS-1588( )/AIC (Assembled)			1.0%	4.0%
Operational Self Check	3.13.1	4.9.1	* 1.0% for the combined group	
Posts, binding (J5) and (J6)	3.13.2	4.9.2		
Posts, binding (J2) and (J3)	3.13.3	4.9.3		
Post, binding (J1)	3.13.4	4.9.4		
Post, binding (J4)	3.13.5	4.9.5		
Operational inspection	3.15	4.12	0.1%	

*All electrical defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table IV and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have met the Group A inspection.

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TABLE IV - GROUP B INSPECTION

Inspection	Req Para	Insp Para	AQL
Jack (J7)	3.13.6	4.9.6	6.5%
Interchangeability	3.16	4.11	6.5%

4.5.3 Group C inspection. - This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.

TABLE V - GROUP C INSPECTION

Inspection	Req Para	Insp Para
Subgroup I		
Altitude	3.12.3	4.8.3
Temperature	3.12.1	4.8.1
Vibration	3.12.6	4.8.6
Humidity	3.12.2	4.8.2
Subgroup II		
Spraytight protection	3.12.4	4.8.4
Shock, bench handling	3.12.7	4.8.7.1
Shock, Drop	3.12.7	4.8.7.2
Subgroup III		
Fungus	3.12.5	4.8.5

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4.5.4 Sampling for inspection of equipment. - Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.4.1 Subgroup I. - For this subgroup, two (2) units from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent Group C inspections, one unit from each successive twenty-five (25) or fraction thereof shall be selected.

4.5.4.2 Subgroup II. - For this subgroup, two units from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, one (1) unit from each successive fifty (50) or fraction thereof shall be selected.

4.5.4.3 Subgroup III. - For this subgroup, only one (1) unit from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent inspection, one (1) unit shall be selected when one half of the units on order have been accepted by the Government's QA Representative.

4.5.5 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective (See 6.4).

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

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4.6 Bounce preconditioning. - Each test set(s) shall be placed in its normal operating position on a L.A.B. ARV-30-200 vibration table. The unit shall be hard mounted to the vibration table and subjected to a two (2) minute vibration. The vibration force shall be at least 1g at 25 Hz. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned or otherwise changed prior to subjection to acceptance inspection.

4.7 Standard test conditions. - Unless otherwise specified, all tests shall be performed at normal room temperature, pressure, and humidity as defined in MIL-STD-810.

4.8 Service conditions. - The service condition tests shall be conducted in accordance with MIL-STD-810 and the test methods reflected in 4.8.1 thru 4.8.7. When any service condition requires a performance test, the equipment shall meet the requirements of 3.13. Upon completion of each service condition test, the equipment shall be subjected to and pass a visual and mechanical inspection. No defects will be allowed.

4.8.1 Temperature. -

4.8.1.1 Operating. - Test Methods 501 and 502 and the limits of 3.12.1.1.

4.8.1.2 Non-operating. - Test Methods 501 and 502 and the limits of 3.12.1.2.

4.8.2 Humidity. - Test Method 507, Procedure III, and the limits of 3.12.2.

4.8.3 Altitude. - Test Method 500, Procedure I, and perform test to comply with 3.12.3.

4.8.4 Spray-tight protection. - Test Method 506 and the parameters of 3.12.4.

4.8.5 Fungus. - The unit(s), including accessories, shall be subjected to 28 days of the fungus test specified in Method 508. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The unit(s) shall be subjected to the test in this open condition and as specified for compliance with 3.12.5 and shall not be required to operate.

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4.8.6 Vibration. - Test Method 514. 1-7 (Curve AY) and the conditions of 3.12.6.

4.8.7 Shock. - Test Method 516, and the conditions of 3.12.7.

4.8.7.1 Shock, bench handling. - Procedure V of Test Method 516.

4.8.7.2 Shock, drop. - Procedure II of Test Method 516.

4.9 Electrical (See 3.13). -

4.9.1 Operational self-check. - Apply a normal 27.5 ± 0.5 volts dc to the test set and Power Off switch S7 in Power Position. Perform the functions as indicated in Table II. The requirement of 3.13.1 shall be met.

4.9.2 Posts, binding (J5) and (J6). - Connect multimeter probes to +28V (J5) and GND (J6) binding posts and set POWER-OFF switch (S7) to POWER. The requirements of 3.13.2 shall be met.

4.9.3 Posts, binding (J2) and (J3). - Connect multimeter probes to +28V (J5) and GND (J6) binding posts and set POWER OFF switch (S7) to POWER. The requirements of 3.13.3 shall be met.

4.9.4 Post, binding (J1). - Ensure no power is applied to the test set. Connect multimeter probes to pin 30 of connector (P1) and AUDIO OSC binding post (J1). The requirement of 3.13.4 shall be met.

4.9.5 Post, binding (J4). - Ensure no power is applied to the test set. Connect multimeter probes to pin 5 of connector (P1) and OUTPUT MTR binding post (J4). The requirement of 3.13.5 shall be met.

4.9.6 Jack (J7). - Ensure no power is applied to the test set. Insert plug (U-174/U) into HDST MIC Jack (J7). Connect one lead of multimeter to pins 3, 5 (S6 in HD Mic position), 22 and 23 of connector (P1), and other to pins 3, 2, 1 and 4 respectively of jack. The requirement of 3.13.6 shall be met.

4.10 Visual and mechanical inspection. - Prior to and after assembly, the equipment shall be examined for defects in accordance with MIL-STD-252, and the requirements of 3.18 shall be met.

4.11 Interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.16. When a listed dimension is not within specified or designed limits, it shall be considered a major defect.

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a. External and internal dimensions of cases, covers and insertable assemblies when such dimensions affect mating of parts.

b. Dimensions of cavities, of the console and other units, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts of assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

4.12 Operational inspection. - Each TS-1588( )/AIC shall be inspected and tested for assurance that all accessories are in place and that the test set controls perform their required functions for compliance with 3.15 when a sample of 20 switch functions are randomly selected from Table II.

4.13 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the packaging requirements of Section 5.

4.14 Rough handling test (preparation for delivery). - When rough handling test, in accordance with MIL-P-116, is required by the contract, the operational inspection test 4.12 shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C, as specified (See 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Test Set, Radio TS-1588( )/AIC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Test Set, Radio TS-1588( )/AIC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

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5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method TC-1.

5.1.1.4.2 Test Set, Radio TS-1588( )/AIC. - Each Test Set, Radio TS-1588 ( )/AIC shall be individually packaged Method III as follows: Cushion the unit on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320 Type CF, Class weather resistant Variety SW, grade W5C, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5C. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specifications.

5.1.2 Level C. - Test Set, Radio TS-1588( )/AIC, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (See 6.2).

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of Test Set, Radio TS-1588( )/AIC, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147, except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3.

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When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto, except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

5.2.2.1 Palletized load. - A quantity of Test Set, Radio TS-1588( )/AIC, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than pelletized load. - When quantities per destination are less than a pallet load, a quantity of Test Set, Radio TS-1588( )/AIC, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Style E, or PPP-B-636, Type CF, Class Weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C. -

5.2.3.1 Palletized load. - A quantity of Test Set, Radio TS-1588( )/AIC, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Test Set, Radio TS-1588( )/AIC, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2, except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - Test Set, Radio TS-1588( )/AIC, in conjunction with standard test equipment is used as an aid for verifying the serviceability of a Control, Intercommunication Set C-1611( )/AIC at direct, general and depot support levels of maintenance.

6.2 Ordering data. - Procurement documents should specify the following and any amendment thereto:

a. Title, number and date of this specification.

b. Level of packaging and level of packing required for shipment (Level A, B or C). See Section 5.

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection:

(1) Several samples of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on samples while complete performance measurements can be made on one sample unit (See 3.3).

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon (including rough handling and functional tests.)

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

i. When rough handling and functional tests are required.

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6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, TS-1588A/AIC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Group C inspection. - Approval to ship may be withheld at the discretion of the Government, pending results of corrective action specified in the contract.

6.5 Definition of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less, and distribution within the continental limits of the United States.

6.6 Location of operational inspection. - It is desirable that the operational inspection 4.12 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.7 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's Quality Control System and the quality history of the product.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-55369(EL) TEST SET, RADIO TS-1588( )/AIC

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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